

Sparse Feature Unlearning Mitigates State Hallucination in Vision-Language-Action Models

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Abstract: Vision-Language-Action models (VLAs) have shown strong generalization in robotic manipulation by leveraging rich representations from pre-trained vision-language models. However, their deployment in real-world environments remains limited by unreliable behaviors. In this work, we study a critical failure mode in VLAs, which we refer to as **state hallucination**, the phenomenon in which a VLA generates actions based on an incorrect implicit belief about the robot-object state. We show that state hallucination is a primary cause of task failures, occurring in 70% of rollouts on LIBERO-Plus. To diagnose this failure mode, we find that state hallucination coincides with weakened attention on task-relevant visual regions, and a mechanistic interpretation via sparse autoencoders reveals that hallucination-relevant sparse features emerge and are activated when these failures occur. Based on this, we propose region-guided feature policy unlearning, which selectively suppresses hallucination-associated region-features while retaining features necessary for successful execution. Experiments in simulated and real-world environments show that our method substantially reduces harmful hallucination and improves task success rates without compromising the original capability. These results suggest that mechanistic feature analysis provides a practical basis for selectively unlearning undesirable behaviors in embodied foundation models.

Keywords: Vision-Language-Action Models, Machine Unlearning, Sparse Autoencoders

1 Introduction

Vision-language-action models (VLAs) have become a promising approach for bringing foundation models into robot manipulation [1, 2, 3]. By integrating language-conditioned task reasoning with visual grounding and low-level action generation, these models enable translation of high-level instructions into continuous robot control. Their success suggests that robot policies can benefit from multi-modal pretrained representations and large-scale heterogeneous robot data, enabling broader generalization across tasks, objects, and embodiments [4, 5, 6]. However, as VLAs move toward real-world deployment, their reliability becomes a central concern [7, 8, 9]. Unlike hallucination in text or vision-language generation, hallucinated actions in VLAs directly affect physical execution and can lead to infeasible or unsafe interactions with the environment [9].

In this work, we investigate **state hallucination**, a critical failure mode of VLAs in which the policy generates actions based on an incorrect implicit belief about the robot-object state. For instance, a VLA may close its gripper before reaching the object and then continue transport and place motions as if the grasp had succeeded, producing a sequence of actions inconsistent with the actual scene (Figure 2). Our analysis on OpenVLA [2] with LIBERO-Plus [7] shows that state hallucination is *prevalent*, occurring in 70% of rollouts and being harmful in 60%. Such hallucination rollouts show weakened attention on task-relevant regions such as the gripper and target object and more scattered attention over the background. Through a mechanistic analysis with sparse autoencoders (SAEs), we

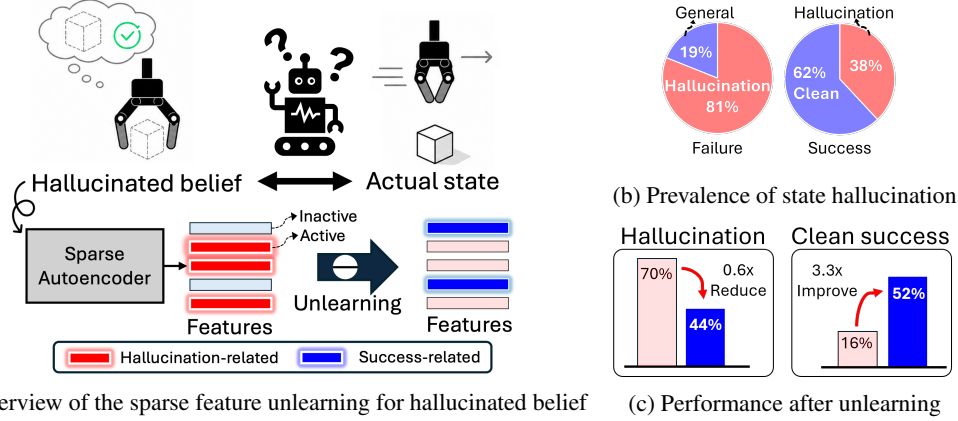


Figure 1: **Sparse feature unlearning mitigates hallucinated beliefs in VLAs.** (a) State hallucination occurs when VLA mistakes its hallucinated belief for the actual robot-object state and generates actions based on this false belief. Hallucination-related features are activated during this mismatch, which are then suppressed through the proposed unlearning while preserving success-related features. (b) Hallucination is a dominant failure mode, accounting for 81% of failures, and remains frequent even in successful trials at 38%. (c) Our proposed sparse feature unlearning reduces hallucination from 70% to 44% and improves clean success without hallucination from 16% to 52%, showing that targeted feature suppression can mitigate state hallucination while improving task completion.

39 further identify hallucination-relevant sparse features that emerge and activate when these failures
 40 occur. These findings establish state hallucination as a diagnosable failure mode of VLAs.

41 Based on this analysis, we propose *Region-Guided Feature Policy Unlearning*, which selects sparse
 42 features that are strongly associated with hallucination in specific visual regions and suppresses their
 43 activations while preserving normal behaviors. Specifically, we identify region-feature pairs whose
 44 activation correlates with hallucination failures, and optimize a feature-level unlearning objective
 45 that suppresses these features without erasing the general manipulation capability of the VLA.
 46 Experiments on LIBERO-Plus and a real-world Franka robot show that our method substantially
 47 reduces harmful hallucination and improves task success while preserving the original manipulation
 48 capability, suggesting that mechanistic feature analysis provides a practical basis for selectively
 49 unlearning undesirable behaviors in embodied foundation models.

50 2 Related Work

51 2.1 Vision-Language-Action Models

52 Vision-language-action models have recently become a major direction for generalist robot ma-
 53 nipulation. One major line of work improves generalization by scaling pretrained vision-language
 54 representations, robot demonstration data, and action-generation architectures [1, 2, 3, 4, 6, 10, 11, 12].
 55 Another line of work improves deployment reliability by adding reasoning, recovery, or test-time
 56 selection: visual reasoning methods predict intermediate cues or goals [13, 14], recovery methods
 57 detect and correct execution failures [15, 16], and test-time scaling methods sample or verify multiple
 58 candidate actions [8, 17]. These approaches improve how actions are generated, corrected, or selected
 59 during inference, while remaining the internal features associated with hallucination-prone behavior
 60 in VLAs. In contrast, our work identifies sparse internal features associated with state hallucination
 61 in specific visual regions and updates the policy to suppress them.

62 2.2 Machine Unlearning

63 Machine unlearning aims to make a trained model forget the knowledge associated with selected
 64 data without retraining the model from scratch [18, 19, 20]. Prior work has studied this problem
 65 mainly in classification tasks, where selected examples are removed, and in language-model tasks,

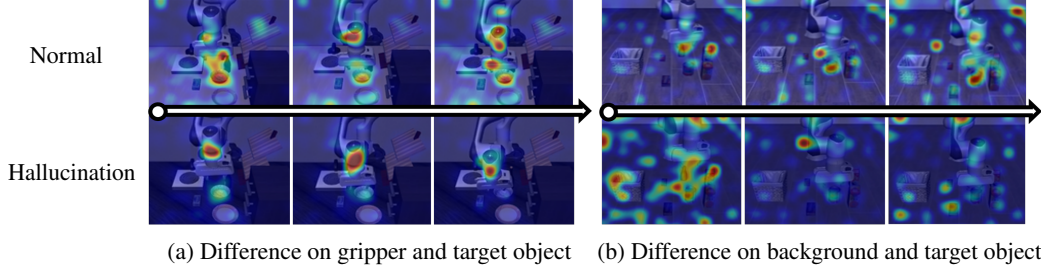


Figure 2: **What is state hallucination?** State hallucination refers to the phenomenon in which a VLA generates actions based on a predicted robot–object state that differs from the actual one, leading to actions inconsistent with the true state. We illustrate it with two representative cases comparing normal (top) and hallucination (bottom) executions under the same task instruction. In both cases, state hallucination coincides with a visible breakdown of attention on task-relevant regions.

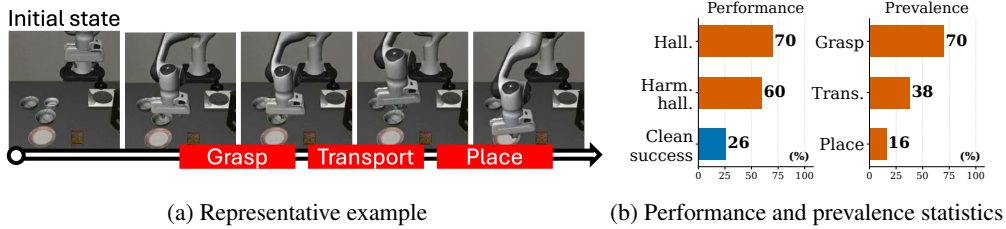


Figure 3: **How does state hallucination appear in VLA execution?** (a) The robot closes the gripper while approaching, transports as though it were securely grasped, and places even though the object is not actually held. (b) Performance and prevalence statistics show frequent state hallucination, with grasp being the most prevalent type. ‘Hall.’ and ‘Trans.’ denote state hallucination and transport.

where factual knowledge or undesired responses are suppressed [21, 22]. A central challenge in machine unlearning is the trade-off between removing undesired knowledge and preserving useful model capabilities. This trade-off is especially important for robot action policies, where undesired behaviors can share perception, language grounding, and motor-control representations with normal skills. This motivates selective policy unlearning methods that suppress representations associated with harmful behaviors while preserving representations needed for successful task execution. In this work, we study policy unlearning for VLAs by selectively suppressing internal features associated with state hallucination while preserving clean task-execution capability.

2.3 Sparse Autoencoders

Sparse autoencoders (SAEs) provide a way to analyze high-dimensional model activations by decomposing dense hidden representations into sparse feature activations [23, 24]. Each sparse feature is activated only for a subset of inputs, making it easier to associate internal model activity with specific inputs or behaviors. Top- k SAEs are widely used because they enforce sparsity with a simple and explicit constraint: only the k largest latent activations are retained and the remaining activations are set to zero for each input [24, 25]. In this work, we employ SAEs to analyze and mitigate state hallucination in VLAs. Specifically, we identify hallucination-associated features across visual regions, and suppress the corresponding feature activations through region-guided policy unlearning.

3 State Hallucination Drives Task Failures

In this section, we show that state hallucination is a primary cause of task failures in current VLAs and analyze where it manifests internally. We first define state hallucination (§3.1), then quantify its prevalence (§3.2), compare attention patterns between normal and hallucination executions (§3.3), and finally probe the internal features that trigger it through sparse autoencoders (§3.4).

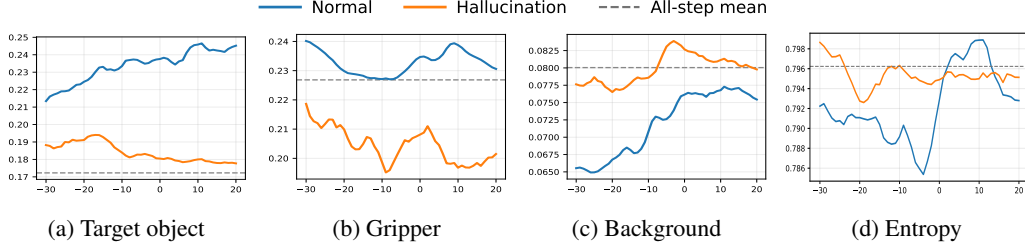


Figure 4: **Quantitative comparison of event-aligned attention patterns between normal and hallucination executions.** The gray dashed line indicates the mean values across all steps. The x-axis is aligned to the corresponding event, where $x = 0$ denotes the time step at which the event occurs.

3.1 What is state hallucination?

We define **state hallucination** as the phenomenon in which a VLA model generates actions that presuppose a robot-object state different from the actual one. Figure 2 illustrates this phenomenon with two representative cases, each comparing a normal and a hallucination execution under the same task instruction. In the left figure, normal execution maintains strong attention on the gripper and target object before the grasp event. In contrast, the hallucination execution shows weaker attention on these task-relevant regions. In the right figure, the model executes a misdirected action toward a non-target region; its attention is scattered over the background and non-target objects, in contrast to the normal execution that concentrates on the gripper and the target object. Across both cases, state hallucination coincides with a visible breakdown in attention to task-relevant regions, suggesting that attention failure is a behavioral pattern of state hallucination.

3.2 State hallucination is a major cause of task failures

To quantify the significance of state hallucination in task failures, we analyze the rollouts of OpenVLA [2] on LIBERO-Plus [7] and measure how frequently state hallucination occurs in Figure 3. Figure 3a illustrates the progress of states during robot manipulation. As shown in Figure 3b, state hallucination appears in 70% of all rollouts, and 60% are harmful hallucinations that directly result in task failure, while only 26% achieve clean success. To further characterize when hallucination occurs, we categorize harmful hallucinations by the manipulation events, which grasp hallucination is the dominant mode, followed by transport and place. These results confirm that state hallucination is not a rare failure mode but a dominant cause of task failure in current VLAs.

3.3 Comparative analyses of normal and state hallucinated manipulations

We quantify attention from the normalized maps using two metrics. First, we measure region-level attention by averaging the normalized attention values over the visual tokens assigned to each semantic region. Second, we measure attention entropy to quantify how spatially dispersed the attention is over visual tokens. As plotted in Figure 4, normal executions assign higher attention to task-relevant regions, especially the target object and gripper. In contrast, hallucination executions assign lower attention on these regions, while assigning relatively higher attention to the background. In terms of attention entropy, normal executions show lower than mean value before the event, indicating more concentrated attention. In contrast, hallucination executions remain near the mean entropy level, suggesting weaker event-related attention concentration.

3.4 Mechanistic interpretation of what triggers state hallucination

To investigate the internal mechanisms underlying state hallucination, we scope the triggering robot representations. Specifically, we decompose the representations of VLA into sparse and interpretable representations using Sparse Autoencoder (SAE) [23], which allows us to identify the specific features associated with hallucination behavior.

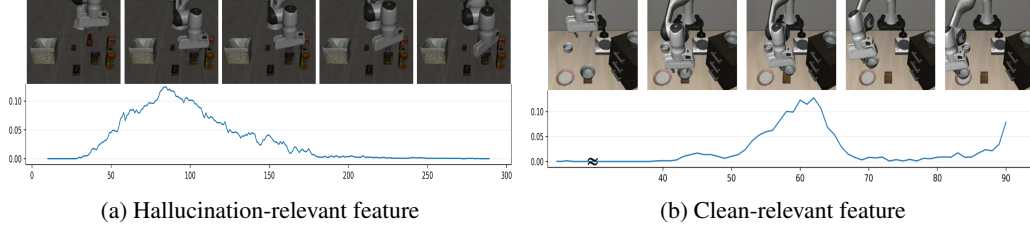


Figure 5: **State hallucination-relevant sparse features exist.** We visualize representative sparse feature activations over rollout trajectories. (a) A hallucination-relevant feature is strongly activated when grasp hallucination occurs. (b) A clean-relevant feature is activated when the policy successfully executes the corresponding subtask and completes the overall task.

SAE training details. We use Top- k SAEs [24, 25] to encode activations into sparse feature representations. To train these SAEs, we construct an activation dataset from rollouts under three outcome conditions: clean success (CS), hallucination failure (HF), and non-hallucination failure (NF), allowing us to distinguish hallucination-specific features from those related to clean execution or general failure. Each SAE learns to reconstruct an activation from a sparse representation, where only the k largest latent activations are retained and the remaining values are set to zero. After training, we freeze the SAE and use its encoder to map policy activations into sparse feature activations.

Hallucination score derivation. Motivated by the region-level attention differences observed in Section 3.3, we condition SAE feature activations on semantic visual regions. For an input sample x , let $\mathcal{R}_r(x)$ denote the set of visual tokens belonging to region r . Let $z_{p,j}(x)$ denote the activation of SAE feature j at visual token p . We define the region-conditioned feature activation as $z_{r,j}(x) = \frac{1}{|\mathcal{R}_r(x)|} \sum_{p \in \mathcal{R}_r(x)} z_{p,j}(x)$. This measures how strongly feature j is expressed in semantic region r .

We next quantify how each region-conditioned feature is expressed across execution outcomes. Since hallucination-related features may be activated only within short event windows, we summarize each rollout by the maximum activation of each region-feature pair over time, $m_{e,r,j} = \max_t z_{e,t,r,j}$. We then average this rollout-level activation within each execution outcome:

$$\bar{m}_{c,r,j} = \frac{1}{|\mathcal{E}_c|} \sum_{e \in \mathcal{E}_c} m_{e,r,j}, \quad c \in \{\text{CS}, \text{NF}, \text{HF}\}. \quad (1)$$

These group-wise activations quantify how strongly each region-feature pair is expressed across the three groups. For each pair (r, j) , we define two complementary scores, namely a hallucination-specific score $s_{r,j}^{\text{hall}}$ and a clean-specific score $s_{r,j}^{\text{clean}}$:

$$s_{r,j}^{\text{hall}} = \bar{m}_{\text{HF},r,j} - \frac{1}{2} (\bar{m}_{\text{CS},r,j} + \bar{m}_{\text{NF},r,j}), \quad s_{r,j}^{\text{clean}} = \bar{m}_{\text{CS},r,j} - \frac{1}{2} (\bar{m}_{\text{HF},r,j} + \bar{m}_{\text{NF},r,j}). \quad (2)$$

Here, a larger $s_{r,j}^{\text{hall}}$ indicates that feature j at region r is selectively activated in hallucination failures compared with the other groups. Conversely, a larger $s_{r,j}^{\text{clean}}$ indicates stronger specificity to clean successful executions. We select the top-ranked features by $s_{r,j}^{\text{hall}}$ and $s_{r,j}^{\text{clean}}$ as hallucination-relevant and clean-relevant features, respectively.

Sparse feature analysis. Figure 5 illustrates how these selected features appear in rollouts: the hallucination-relevant feature becomes strongly activated around a grasp hallucination, whereas the clean-relevant feature is activated during successful subtask execution and task completion. These results indicate that sparse features capture behaviorally meaningful signals associated with hallucination and clean execution.

4 Proposed Method: Region-Guided Feature Policy Unlearning

Our goal is to reduce state hallucination while preserving normal behaviors. The analysis in Section 3 suggests that hallucination is associated with changes in visual grounding over semantic regions and

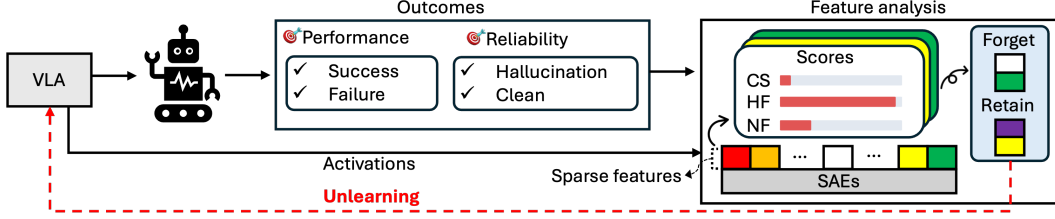


Figure 6: **Overall framework of the proposed method.** First, we identify visual-region-specific sparse features associated with hallucination and clean rollouts from VLA activations. Second, we unlearn the policy with a feature-level unlearning objective that suppresses the selected hallucination-associated features while preserving clean-associated features needed for normal behaviors.

155 sparse internal features. Based on this analysis, we propose *Region-Guided Feature Policy Unlearning*,
 156 which selectively suppresses hallucination-associated sparse features grounded in specific visual
 157 regions, as illustrated in Figure 6.

158 **Region-guided feature selection for forgetting and retention.** We construct feature sets for
 159 forgetting and retention based on the hallucination-specific and clean-specific scores defined in
 160 Section 3.4. To further isolate features specific to hallucination failures, we introduce a separation
 161 score as $s_{r,j}^{\text{sep}} = \bar{m}_{\text{HF},r,j} - \bar{m}_{\text{NF},r,j}$, which measures specificity to hallucination failures over general
 162 failures. For the forget set $\mathcal{F}$, we rank region-feature pairs using an aggregated score $s_{r,j}^{\mathcal{F}} =$
 163 $s_{r,j}^{\text{hall}} + s_{r,j}^{\text{sep}} - s_{r,j}^{\text{clean}}$. This score prioritizes features that are active in hallucination failures, distinct
 164 from non-hallucination failures, and weakly associated with clean execution. We select the top-ranked
 165 pairs as the forget set $\mathcal{F}$. For the retain set $\mathcal{G}$, we select region-feature pairs with high $s_{r,j}^{\text{clean}}$. The
 166 resulting sets guide unlearning to suppress state-hallucination-associated features while preserving
 167 features needed for clean manipulation.

168 **Feature-guided policy unlearning.** After selecting the forget set $\mathcal{F}$ and retain set $\mathcal{G}$, we update the
 169 VLA to suppress hallucination-associated features while preserving clean-associated features. During
 170 unlearning, the SAE encoders are frozen, and only the VLA parameters are updated with LoRA [26].

171 For hallucination samples, the forget loss penalizes the selected forget pairs, while for clean successful
 172 samples, the retain loss encourages the selected retain pairs to remain active:

$$\mathcal{L}_{\text{forget}} = \mathbb{E}_{x \in \mathcal{D}_{\text{HF}}} \left[\frac{1}{|\mathcal{F}|} \sum_{(r,j) \in \mathcal{F}} (z_{r,j}(x; \theta))^2 \right], \mathcal{L}_{\text{retain}} = -\mathbb{E}_{x \in \mathcal{D}_{\text{CS}}} \left[\frac{1}{|\mathcal{G}|} \sum_{(r,j) \in \mathcal{G}} z_{r,j}(x; \theta) \right]. \quad (3)$$

173 Combining the forgetting and retention terms, the final unlearning objective is defined as $\mathcal{L}_{\text{unlearn}} =$
 174 $\mathcal{L}_{\text{forget}} + \lambda_r \mathcal{L}_{\text{retain}}$, where λ_r controls the strength of retention.

175 5 Experiments

176 **Setup.** We conduct evaluation on LIBERO-Plus [7] using OpenVLA [2], a widely used and represen-
 177 tative autoregressive model. LIBERO-Plus extends LIBERO [27] with realistic visual perturbations
 178 and multiple severity levels, making it suitable for evaluating execution reliability. To provide a
 179 reliable reference for comparing normal and hallucination executions, we select tasks on which the
 180 original policy achieves stable success in the corresponding original LIBERO. Specifically, we use
 181 representative task instructions from the LIBERO-Plus suites: *put the bowl on the stove* from the
 182 goal suite, *pick up the black bowl on the cookie box and place it on the plate* from the spatial suite,
 183 and *pick up the tomato sauce and place it in the basket* from the object suite. More implementation
 184 details are provided in the supplementary materials.

185 **Metrics.** We report task-level metrics, including overall success and clean success, and hallucination-
 186 level metrics, including hallucination and harmful hallucination. Overall success denotes task

Table 1: Performance comparison in terms of hallucination and task success.

Method	Hallucination ↓	Harmful hallucination ↓	Overall success ↑	Clean success ↑	Grasp ↓	Transport ↓	Place ↓
Original Policy	70.0	60.0	26.0	16.0	70.0	38.0	16.0
Gradient Ascent	84.0	84.0	0.0	0.0	84.0	62.0	0.0
Region-Guided Feature	44.0	38.0	58.0	52.0	44.0	20.0	6.0

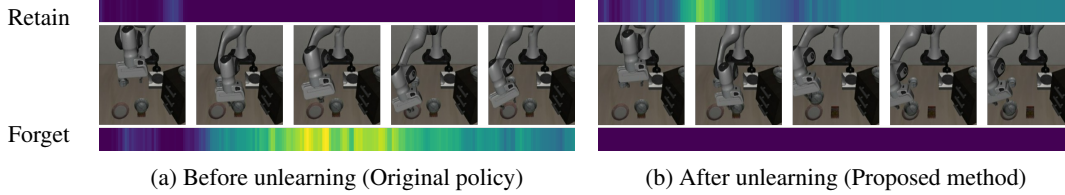


Figure 7: Comparison of rollouts and selected forget and retain feature activations before and after unlearning in benchmark.

completion, while clean success denotes task completion without state hallucination. Hallucination denotes executions with at least one state hallucination event, and harmful hallucination denotes hallucinated executions that fail the task. We also report event-level hallucination rates.

5.1 Results

Does the proposed region-guided feature unlearning improve reliability? As in Table 1, the original policy shows frequent hallucination with 70%, while achieving only 16% clean success. Naive gradient ascent, which directly increases the loss on hallucination samples, further degrades reliability, increasing hallucination to 84% and reducing both success metrics to 0%. This suggests that naive gradient ascent disrupts the manipulation policy instead of selectively removing hallucination-related behavior. In contrast, our method reduces harmful hallucination to 38%, while improving clean success 52%. These results show that the proposed method mitigates hallucination-related failures without destroying useful manipulation capabilities for reliability.

Does unlearning suppress hallucination-related features while recovering clean-relevant features? As in Figure 7, the original policy fails with state hallucination. The selected forget features are strongly activated, whereas the retain features remain nearly inactive. After applying our method, the policy completes the task without hallucination. The forget features are largely suppressed, while the retain features become active during successful manipulation. These results show that our method suppresses hallucination-related activations while recovering clean task-relevant activations.

Does region guidance improve feature unlearning? To evaluate the role of spatial grounding in feature selection, we report ablations with different spatial granularities in Table 2. Global feature selection provides limited gains, suggesting that pooling over all visual tokens can lose spatially localized failure signals. Tokenized feature selection improves over the global variant, indicating that preserving visual-token-level information helps feature selection. Region-guided feature selection achieves the strongest overall improvement across hallucination and success metrics. These results indicate that semantic region guidance enables more selective feature unlearning by suppressing hallucination-related features while preserving clean manipulation behavior.

Does the proposed method preserve clean successful behavior while improving hallucination failures? Figure 8 reports the unlearning trade-off in terms of preservation and improvement. The proposed method preserves most clean behavior, retaining 85% of originally successful task executions. It also improves previously failed behavior, converting 49% of failed executions into successful ones and suppressing hallucinated events up to 88%. These results show a favorable unlearning trade-off, reducing hallucination-related failures while largely preserving clean behaviors.

Do score aggregation and the retain objective improve reliability? Table 3 ablates score aggregation and the retain objective. Removing the retain objective still reduces hallucination, but clean success decreases compared with the full method. Without score aggregation in third row, both

Table 2: **Comparison between global, tokenized, and region-guided feature unlearning.** Global Feature pools activations across all visual tokens, Tokenized Feature selects features at the visual-token level, and Region-Guided Feature selects features using semantic-region-conditioned activations. Values in parentheses denote changes relative to the perturbation baseline.

Method	Hallucination ↓	Harmful hallucination ↓	Overall success ↑	Clean success ↑
Global Feature	62.0 (-8.0)	56.0 (-4.0)	34.0 (+8.0)	28.0 (+12.0)
Tokenized Feature	60.0 (-10.0)	50.0 (-10.0)	44.0 (+18.0)	34.0 (+18.0)
Region-Guided Feature	44.0 (-26.0)	38.0 (-22.0)	58.0 (+32.0)	52.0 (+36.0)

Table 3: **Ablation study of score aggregation and the retain objective.** The aggregated score $s_{r,j}^{\text{agg}}$ combines $s_{r,j}^{\text{sep}}$ and $-s_{r,j}^{\text{clean}}$ for forget target selection.

$s_{r,j}^{\text{forget}}$	$s_{r,j}^{\text{agg}}$	$\mathcal{L}_{\text{retain}}$	Hallucination ↓	Harmful hallucination ↓	Overall success ↑	Clean success ↑
✓	✓	✓	44.0 (-26.0)	38.0 (-22.0)	58.0 (+32.0)	52.0 (+36.0)
✓	✓	✗	46.0 (-24.0)	36.0 (-24.0)	58.0 (+32.0)	48.0 (+32.0)
✓	✗	✓	54.0 (-16.0)	50.0 (-10.0)	48.0 (+22.0)	44.0 (+28.0)

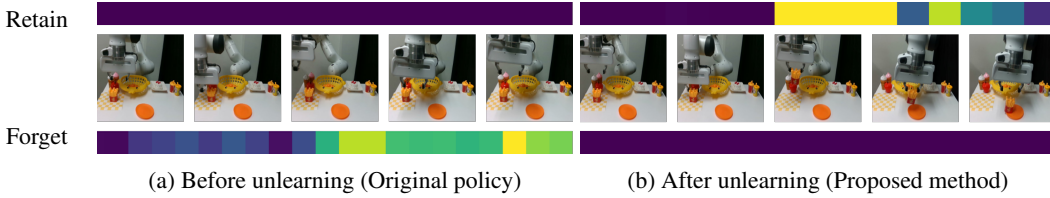


Figure 9: **Real-world comparison of rollouts and feature activations before and after unlearning with a Franka robot with the the instruction “Pick fries and put it on the plate”.**

222 hallucination reduction and task success become weaker, showing that the raw forget score alone is
 223 less selective. Combining score aggregation with the retain objective achieves the best balance across
 224 hallucination and success metrics. These results show that score aggregation improves the selectivity
 225 of forget target selection, while the retain objective helps preserve clean manipulation behavior.

226 **Does the proposed method remain effective in real-world environments?** Figure 9 shows that the
 227 original policy fails with grasp, transport, and place hallucination, accompanied by strong activation
 228 of the selected forget features and weak activation of the retain features. After applying our method,
 229 the robot completes the task without hallucination, while the forget features are suppressed and the
 230 retain features become active. Quantitatively, clean success increases from 24% to 36%, and harmful
 231 hallucination decreases from 52% to 40%. These results suggest that the proposed policy unlearning
 232 can suppress hallucination-related behavior and improve reliability in physical robot execution.

233 6 Conclusion

234 We have presented a study of state hallucination in VLAs, where a policy generates actions based on
 235 an incorrect implicit belief about the robot-object state. Our analysis showed that state hallucination
 236 is reflected in weakened attention to task-relevant visual regions and in hallucination-relevant sparse
 237 features identified by SAEs. Based on these findings, we proposed Region-Guided Feature Policy
 238 Unlearning, which selectively suppresses hallucination-associated region-feature pairs while retaining
 239 features needed for successful execution. Experiments in simulated and real-world environments
 240 show that our method reduces harmful hallucination and improves task success without compromising
 241 the original manipulation capability. These results suggest that mechanistic feature analysis provides
 242 a practical basis for diagnosing and unlearning undesirable behaviors, offering a step toward more
 243 interpretable and reliable VLA-based robot manipulation policies.

244 **Limitations.** Although this work identifies state hallucination and reduces it through feature unlearn-
 245 ing, evaluation is limited to pick-and-place, which represent a fundamental class of manipulation
 246 behaviors. Future work can further examine the proposed framework on broader manipulation
 247 benchmarks and more diverse real-world settings, including long-horizon and bimanual tasks.

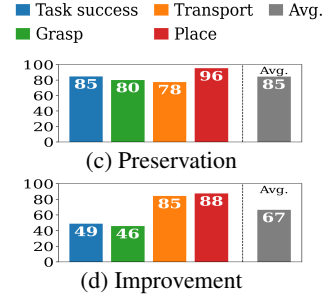


Figure 8: **Unlearning trade-offs.** Preservation measures retained correct behaviors, while improvement measures failures corrected after unlearning.

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